EE2211 Midterm AY2324 Sem1

Introduction to Machine Learning (National University of Singapore)

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# of Questions: 22

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Question #: 1

This question is related to the understanding of linear systems and partial derivatives. Which of the following statements below is/are correct?

unknowns > eqn

I. In under-determined linear systems, the number of parameters is greater than the number of unknown equations.

eqn < unknowns

II. The system

$$\begin{bmatrix} 1 & 4 \\ 2 & 7 \\ -3 & 11 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -2.5 \\ 4 \end{bmatrix}$$

3 eqn 2 unknowns
overdetermined system, approx sol'n
= left inv

has no *exact* solution but an approximated solution is available using the left inverse.

III. If **f**(**x**) is a vector-valued function of *p outputs* and **x** is a 2-dimensional vector, then differentiation of **f**(**x**) with respect to **x** is an 2 *x p* matrix.

IV. Consider the linear system

$$\mathbf{X}\mathbf{w} = \mathbf{y}, \mathbf{X} \in R^{m \times d}$$

m = no. of linear eqn
d = no. of parameters / unknowns

is the input data matrix,

$$\mathbf{w} \in R^{d \times 1}$$

is the parameter vector, and

$$\mathbf{y} \in R^{m \times 1}$$

is the target vector. If d >m, the system has ~~more~~ equations than parameters.

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✓ I & II
I & IV
I, II & III
II, III & IV
None of the others.

Item Weight: 3.0

Question #: 2

statistical

Causality is a ~~deterministic~~ relationship; suppose we know A and B have causal relation, if A occurs, B is for sure to take place.

A. True
✓B. False

Item Weight: 2.0

Question #: 3

used to calculate probability of events that cannot occur together

The sum rule of probability enables us to transform conditional probability to joint probability.

only used in diff scenarios

given that one event has already occured

A. True
✓B. False

Item Weight: 2.0

Question #: 4

not the best approach

It is a common practice to replace missing data (such as age) with an out-of-range value (such as -1), so as to estimate the mean and standard deviation of the data.

A. True
✓B. False

Item Weight: 2.0

Question #: 5

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all info provided labelled data

We use an unsupervised learning scheme to group the covid cases across United States into geographical clusters. As such, we are performing deductive reasoning to derive the clusters.

rule based reasoning

A. True
✓B. False

Item Weight: 2.0

Question #: 6

no pre-labelled outputs

?

In unsupervised learning, we are given the samples and their corresponding output, and our goal is to discover the underlying pattern of the samples.

A. True
✓B. False

Item Weight: 2.0

Question #: 7

We have an unbiased die with six faces (i.e., 1,2,3,4,5,6), where each face has equal probability to show up in a rolling game.

$6^2 = 36$ total outcomes

Suppose we roll the die twice, and then count the sum of the faces showing up. For example, if the first time 1 shows up and the second time 3 shows up, the sum would be 1+3=4.

even no. sum > 8

What is the probability of obtaining an sum being an even number greater than 8?

$\frac{4}{36} = \frac{1}{9}$

4 6
6 4
5 5
6 6

4 sets of outcomes

A. 1/12

✓B. 1/9

C. 1/36

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D. 1/8
E. 1/6
F. None of others is correct.

Item Weight: 3.0

Question #: 8

Please select the correct option.

A. Classification can be only applied on ratio and interval data, but not on ordinal or nominal data.
B. Clustering can be only applied on ratio and interval data, but not on ordinal or nominal data.
C. The output of a classifier can be only ratio and interval data, but not ordinal or nominal data.
D. (A) and (B)
E. (B) and (C)
F. (A), (B), and (C)
✓G. None of others is correct.

Item Weight: 3.0

Question #: 9

We have a collection of 1,000 images from three classes: cat, bird, and dog. With these images, we would like to train an image classifier that categorizes an input image into one of the three classes.

To ensure the 1,000 images are of good quality for training the classifier, we ask a well-trained human inspector to go through all the images, to label the images and remove noisy ones. Eventually, we revmoved 200 images of low quality suggested by the inspector, and use the remaining 800 images to train the classifier; the 800 images comprise 200 cat images, 300 bird images, and 300 dog images.

Please select the correct option.

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A. The human inspection process can be considered as a data cleaning step.

B. If we are to use one-hot encoding for the labels of the three classes, we can set

Cat = [1 1 1]
Dog = [0 1 0]
Bird = [0 0 0].

all must be 1 diagonally

C. The image classification conducted here is an unsuperivsed-learning task.

D. If we keep the 200 noisy images (suggested by the human inspector), we will end up having more training images and hence a better-performed image classifier.

E. (a) and (b)

F. (a), (b), and (d)

G. None of others is correct.

Item Weight: 3.0

Question #: 10

There are 200 students in a training camp, who are divided into two training groups, 120 students in Group A and 80 students in Group B. Let G be the variable that stands for the group that a student belongs to, which can be either 'a' (Group A) or 'b' (Group B). Hence, we know Pr(G=a)=120/200=0.6.

$Pr(G=b) = 1-0.6$
$= 0.4$

The students take part in a fitness test. Let T be the variable that stands for the result of the fitness test, which can be either 'p' (passing the test) or 'f' (failing the test).

$P(G=b \mid T=f) = 0.52$ b and failed

We know that out of the 200 students, 100 passed the fitness test, i.e., P(T=p) =100/200=0.5; others failed the test. We also know that, for all those who failed the test, 52% comes from Group B, i.e., P(G=b | T=f) =0.52.

$P(G=b \mid T=p) = \frac{P(T=p \mid G=b) \cdot P(G=b)}{P(T=p)}$

What is the probability of students from group A who passed the test, i.e., P(T=p | G=a)?

$= \frac{(0.4)}{0.5} = 0.52$

1

Please write your answers to three decimal places (e.g., 0.111).

$P(T=p \mid G=b) = \frac{P(G=b \mid T=f) \cdot P(T=f)}{Pr(G=b)}$

$= \frac{0.52 \times 0.5}{0.4} = 0.65$

$\boxed{P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B)}}$

Bayes Theorem

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Q10. 200 students — 120 group A, 80 group B

$Pr(G=a) = \frac{120}{200} = 0.6$

$Pr(G=b) = 1-0.6 = 0.4$

$P(T=P) = \frac{100}{200} = 0.5$

$P(T=F) = 1-0.5 = 0.5$

① $Pr(G=b \mid T=f) = 0.52$ given in q1

② $Pr(G=b \mid T=P) = \frac{Pr(T=P \mid G=b) \times P(G=b)}{Pr(T=P)}$

$= \frac{(1-0.65) \times 0.4}{0.5} = 0.28$

③ $P(T=P \mid G=b) = 1 - Pr(T=f \mid G=b)$

$= 1-0.65$

$= 0.35$

④ $Pr(G=a \mid T=P) = 1 - Pr(G=b \mid T=P)$

$= 1-0.28$

$= 0.72$

$Pr(T=f \mid G=b)$

$= \frac{Pr(G=b \mid T=f) \times Pr(T=f)}{Pr(G=b)}$

$= \frac{0.52 \times 0.5}{0.4}$

$= 0.65$ (tail in B)

Bayes Theorem: $P(T=P \mid G=a)$

$= \frac{P(G=a \mid T=P) \times P(T=P)}{P(G=a)}$

$= \frac{0.72 \times 0.5}{0.6}$ given in q2

$= 0.6$ #

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1. Range - Min:0.580 Max:0.620

Item Weight: 4.0

Question #: 11

rank is zero if no linearly independent rows present

Given a matrix

$$\mathbf{X} = \begin{bmatrix} 1 & -2 \\ -4 & a \end{bmatrix}$$

python code chp 4.5

max no. of linearly independent rows

.

We know that rank(**X**)=1, the value of a should be 1

1. Range - Min:7.99999 Max:8.000001

Item Weight: 4.0

Question #: 12

Please select the correct option.

A. We would like to design an image classification system. During inference/testing, we would need to implement a two-step procedure: Step 1 is to ~~conduct classification~~ clean data, and Step 2 is to extract features.

B. Classification is a supervised learning task; we would need to extract discrete features for the samples, since classification works with discrete features ~~but not~~ continuous features.

C. Clustering is an unsupervised learning task; we would need to extract discrete features for the samples, since clustering works with ~~discrete features but not continuous features.~~

D. (A), (B), and (C)

E. (B), and (C)

F. None of others is correct.

Item Weight: 3.0

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Question #: 13

We would like to design a linear regression function to predict future cognitive performance and depressive severity in the elderly. We have collected two clinical features of 5 participants (x_1, x_2) that are related to future cognitive decline and depression. We also have the five-year follow-up data of cognitive scores (1st columne of **Y**) and depressive scores (2nd column of **Y**) of these 5 participants (see below).

tall = overdetermined sys, left inV

$$\mathbf{X} = \begin{bmatrix} 50 & 10 \\ 40 & 7 \\ 65 & 12 \\ 70 & 5 \\ 75 & 4 \end{bmatrix}, \mathbf{Y} = \begin{bmatrix} 9 & 3 \\ 6 & 7 \\ 5 & 6 \\ 3 & 1 \\ 2 & 9 \end{bmatrix}.$$

i) Perform least square estimation based on the existing training data. What is the mean of squared error of the estimated model for cognitive scores? __1__ (up to 4 decimal places) What is the mean of squared error of the estimated model for depressive scores? __2__ (up to 4 decimal places)

ii) If we have a new participant with the two clinical features (x_1, x_2) =(62, 8). What is his/her five year follow-up cognitive score? __3__ (up to 4 decimal places) What is his/her five year follow-up depressive score? __4__ (up to 4 decimal places)

1. Range - Min:1.22 Max:1.23 1.23
2. Range - Min:8.01 Max:8.02 8.02
3. Range - Min:4.91 Max:4.92 4.91
4. Range - Min:5.13 Max:5.14 5.14

Item Weight: 16.0

Question #: 14

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Which of the following is/are true about matrix below?

$$\mathbf{A} = \begin{bmatrix} 1 & 4 & 2 \\ -3 & 10 & 5 \end{bmatrix}$$

rectangular wide, under determined sys
right inv
disregard det unless even det
~0

I) **A** is invertible
II) **A** is left invertible
III) **A** is right invertible ✓
IV) **A** has no determinant ✓

A. I &III
✓B. III &IV
C. II &IV
D. II, III &IV
E. None of the above

Item Weight: 3.0

Question #: 15

Select the correct options regarding derivatives.

I) Suppose g(**x**) is a *scalar* function of *b* variables where **x** is a *bx*1 vector. Without taking data points into consideration, the outcome of differentiation of $\mathbf{g}(\mathbf{x})$ w.r.t. **x** is a ~~scalar~~. vector

II) Suppose $\mathbf{f}(\mathbf{x})$ is a *scalar* function of *d* variables where **x** is a *dx*1 vector. Without taking data points into consideration, the outcome of differentiation of $\mathbf{f}(\mathbf{x})$ w.r.t. **x** is a *dx1* vector. ✓

III) If $\mathbf{f}(\mathbf{x})$ is a vector function of *size bx*1 and **x** is a *d*x1 vector, then differentiation of $\mathbf{f}(\mathbf{x})$ w.r.t. **x** is a *bxd* matrix. ✓

IV) If $\mathbf{f}(\mathbf{x})$ is a vector function of *size bx*1 and **x** is a *d*x1 vector, then differentiation of $\mathbf{f}(\mathbf{x})$ w.r.t. **x** is a ~~*dx1*~~ vector. bxd

A. I &III
B. I &IV
C. II &IV

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✓D. II &III

E. None of the above

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Question #: 16

A set of linear equations is written as

$$\mathbf{w}^T\mathbf{X} = \mathbf{y}^T$$

m= linear eqns (rows)
d= parameters (columns) or unknowns

where

m d

$$\mathbf{X} \in \mathcal{R}^{4\times\boxed{3}}$$

and

$$\mathbf{y} \in \mathcal{R}^{\boxed{3}\times 1}$$

. How many simultaneous equations are there in this set of equations?

A. 1
B. 2
✓C. 3
D. 4
E. 5

Item Weight: 3.0

Question #: 17

The values of feature x and their corresponding values of target y are shown in the table below.

x	3	4	5	6	7
y	5	4.5	4	3.5	3

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chp 5

Based on the least square regression line y =a *x* +b, please select the correct option below.

I) The estimated a* and b* minimize the sum of squared errors between the predicted y' and the target output y for all training samples.

II) The sum of squared errors between the predicted y' and the target output y for all training samples is called the loss function.

III) The estimation of y at *x* =4.5 may be accurate as the prediction range is within the given data range.

IV) The estimation of y at *x* =1000 should be accurate as the prediction is linear.

✓A. I &III

B. III

C. I, II &III

D. I &IV

E. None of the above

Item Weight: 3.0

Question #: 18

A learning problem is attempted by minimizing the sum of squared errors given by

$\sum_{i=1}^{m}(f_{\mathbf{w},b}(\mathbf{x}_i) - \mathrm{y}_i)^2 = (\mathbf{Xw} - \mathbf{y})^T(\mathbf{Xw} - \mathbf{y})$ where $\mathbf{X} \in \mathcal{R}^{10\times 11}$ and $\mathbf{w} \in \mathcal{R}^{11\times 1}$

loss function

denote respectively the data matrix and the parameter vector. The outcome ~~must~~ be a unique solution since the square function is convex.

A. True

✓B. False

Item Weight: 2.0

Question #: 19

This question is related to the determination of types of system where an appropriate solution can be found subsequently. The linear system below has no *exact* solution.

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$$\begin{bmatrix} 1 & 3 & 2 \\ 3 & 6 & 4 \\ 5 & -1 & 6 \\ 9 & 8 & 0 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 9 \\ 3 \end{bmatrix}$$

4 eqn, 3 unknowns
overdetermined sys , no exact soln

✓A. True
B. False

Item Weight: 2.0

Question #: 20

This question is related to determination of types of system where an appropriate solution can be found subsequently. The following matrix has a ~~left~~ right inverse.

$$\mathbf{X} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

rectangular wide
underdetermined sys

A. True
✓B. False

Item Weight: 2.0

Question #: 21

This question is related to understanding of modelling assumptions.

$$f(\mathbf{x}) = 6x_1 + 9x_2 - 5x_3 + 3$$

bias

is an affine function.

✓A. True
B. False

Item Weight: 2.0

Question #: 22

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$T: R^3 \to R^1$
(x, y, z)
3×1

A vector function has the mapping of

$$f(\mathbf{x}): \mathcal{R}^3 \to \mathcal{R}^2$$

. The gradient of this function with respect to **x** is a matrix of partial derivatives, which is a 2×3 matrix.

✓A. True
B. False

Item Weight: 2.0

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